

Assignment II - Discrete Math(H)

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Q.1

Q.1 Suppose that A , B and C are three finite sets. For each of the following, determine whether or not it is true. Explain your answers.

(1) $(A - B = A) \rightarrow (B \subseteq A)$

(2) $(A - B = \emptyset) \rightarrow (A \cap B = B \cap A)$

(3) $(A \subseteq B) \rightarrow (|A \cup B| \geq 2|A|)$

Sol:

1. False.

We have $A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \overline{B}$, thus $A \cap \overline{B} = A$, which is to say, $A \cap B = \emptyset$, thus $B \subseteq A$ obviously can't be implied.

2. True.

Intersection is commutative, thus $A \cap B = B \cap A$ is always true, thus the implication is always true.

3. False.

If $A \subseteq B$, then $A \cup B = B$, thus $|A| \leq |B|$, and $|A| \leq |A \cup B|$. But for $|A \cup B| \geq 2|A|$, it's obviously not generally true. Counterexample: $A = B = \{1\}$.

Q.2

Q.2 Let's formulate the "Barber's paradox" in the language of predicate logic. In English, the paradox may be stated as:

"The barber of the village Seville shaves those residents of Seville who do not shave themselves."

Assume that S is the set of all residents of Seville, which includes the barber. We have the following predicates over elements of the set S :

- $Shaves(x, y)$: true if x shaves y , false otherwise.
- $Barber(x)$: true if x is the barber of Seville (you may assume that Seville has just one barber), false otherwise.

Rewrite the statement of the paradox using only these two predicates, along with the notation of mathematical logic. Please also state the reason why the paradox occurs in the logical statement.

Sol:

$$\exists b \left(Barber(b) \wedge \forall y, Shaves(b, y) \leftrightarrow \neg Shaves(y, y) \right)$$

Substitute that $y = b$, we have $Shaves(b, b) \leftrightarrow \neg Shaves(b, b)$ which is false. Thus the statement will always lead to a contradiction, which is a paradox.

Q.3

Q.3 Prove or disprove the following.

- (1) For any three sets A, B, C , $C - (A \cap B) = (C - A) \cap (C - B)$.
- (2) For any two sets A, B , $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$, where $\mathcal{P}(A)$ denotes the power set of the set A .
- (3) For any two sets A, B , $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$, where $\mathcal{P}(A)$ denotes the power set of the set A .
- (4) For a function $f : X \rightarrow Y$, $f(A \cap B) = f(A) \cap f(B)$, for any two sets $A, B \subseteq X$.

1. False.

$$\text{LHS} = C \cap \overline{A \cap B} = C \cap (\overline{A} \cup \overline{B}) = (C - A) \cup (C - B) \neq \text{RHS}$$

2. True.

Let $a \in \mathcal{P}(A) \cap \mathcal{P}(B)$, then $a \subseteq A$ and $a \subseteq B$, thus $a \subseteq A \cap B$, which represents $a \in \mathcal{P}(A \cap B)$.

Let $a \in \mathcal{P}(A \cap B)$, then $a \subseteq A \cap B$, thus $a \subseteq A$ and $a \subseteq B$, which represents $a \in \mathcal{P}(A) \cap \mathcal{P}(B)$.

3. False.

Some subsets of $A \cup B$ contain elements from both A and B , which do not belong to any of $\mathcal{P}(A)$ or $\mathcal{P}(B)$. Which leads it to be false.

4. False.

If $y \in f(A \cap B)$, there exists $x \in A \cap B$ s.t. $f(x) = y$. Hence $y \in f(A)$ and $y \in f(B)$, which is $y \in f(A) \cap f(B)$.

But if $y \in f(A) \cap f(B)$, then when $\exists a \in A, \exists b \in B, y = f(a) = f(b)$, and when $a \notin B$ and $b \notin A$, then there's no $a, b \in A \cap B$, which represent $y \notin f(A \cap B)$.

Q.4

Q.4 Give an example of two uncountable sets A and B such that the intersection $A \cap B$ is

- (a) finite,
- (b) countably infinite,
- (c) uncountable.

Sol:

1. $A = [0, 1]$, $B = [1, 2]$, then $A \cap B = \{1\}$, which is finite.
2. $A = [0, 1] \cup \mathbb{Z}$, $B = [2, 3] \cup \mathbb{Z}$, then $A \cap B = \mathbb{Z}$, which is countably infinite.
3. $A = [0, 2]$, $B = [1, 3]$, then $A \cap B = [1, 2]$, which is uncountable.

Q.5

Q.5 The *symmetric difference* of A and B , denoted by $A \oplus B$, is the set containing those elements in either A or B , but not in both A and B . Give an example of two uncountable sets A and B such that the intersection $A \oplus B$ is

- (a) finite,
- (b) countably infinite,
- (c) uncountable.

Sol:

1. $A = [0, 1]$, $B = [0, 1] \cup \{2\}$, then $A \oplus B = \{2\}$, which is finite.
2. $A = [-2, -1]$, $B = [-2, -1] \cup \mathbb{N}$, then $A \oplus B = \mathbb{N}$, which is countably infinite.
3. $A = [0, 1]$, $B = [2, 3]$, then $A \oplus B = [0, 1] \cup [2, 3]$, which is uncountable.

Q.6

Q.6 Show that if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$.

PF:

We have $\forall (a, b) \in A \times B, a \in A \wedge b \in B$.

And for $A \subseteq C, B \subseteq D$, we obtain $a \in C, b \in D$, which is $(a, b) \in C \times D$.

Therefore, $A \times B \subseteq C \times D$.

Q.E.D..

Q.7

Q.7 Suppose that f is an invertible function from Y to Z and g is an invertible function from X to Y . Show that the inverse of the composition $f \circ g$ is given by $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

PF:

We have:

$$(f \circ g) \circ (g^{-1} \circ f^{-1}) = f \circ (g \circ g^{-1}) \circ f^{-1} = f \circ I_Y \circ f^{-1} = I_Z$$

and:

$$(g^{-1} \circ f^{-1}) \circ (f \circ g) = g^{-1} \circ (f^{-1} \circ f) \circ g = g^{-1} \circ I_Y \circ g = g^{-1} \circ g = I_X.$$

Thus $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

Q.E.D..

Q.8

Q.8 Suppose that two functions $g : A \rightarrow B$ and $f : B \rightarrow C$ and $f \circ g$ denotes the *composition* function.

- (a) If $f \circ g$ is one-to-one and g is one-to-one, must f be one-to-one? Explain your answer.
- (b) If $f \circ g$ is one-to-one and f is one-to-one, must g be one-to-one? Explain your answer.
- (c) If $f \circ g$ is one-to-one, must g be one-to-one? Explain your answer.
- (d) If $f \circ g$ is onto, must f be onto? Explain your answer.
- (e) If $f \circ g$ is onto, must g be onto? Explain your answer.

Sol:

1. f doesn't have to be one-to-one.

Let $A = \{1, 2\}$, $B = \{a, b, c\}$, $C = \{0, 1\}$, and $g(1) = a, g(2) = b, f(a) = 0, f(b) = 1, f(c) = 1$, then $(f \circ g)(1) = 0, (f \circ g)(2) = 1$, which is an injection.

2. g must be one-to-one.

If $\exists x_1, x_2, s. t. x_1 \neq x_2 \wedge g(x_1) = g(x_2)$, then $(f \circ g)(x_1) = (f \circ g)(x_2)$, leads to a contradiction. Thus g is an injection.

3. g must be one-to-one.

Reason is same as 2.

4. f must be onto.

We have $\forall z \in C, \exists x \in A, (f \circ g)(x) = z$, let $y = g(x)$, then $f(y) = z$, i.e., f is a surjection.

5. g doesn't have to be onto.

Let $A = \{a, b\}$, $B = \{0, 1, 2\}$, $C = \{0, 1\}$, $g(a) = 0, g(b) = 1, f(0) = 0, f(1) = 1, f(2) = 1$, then $(f \circ g)(a) = 0, (f \circ g)(b) = 1$, which is a surjection.

Q.9

Q.9 Derive the formula for $\sum_{k=1}^n k^2$.

Sol:

$$\begin{aligned}k^3 - (k-1)^3 &= 3k^2 - 3k + 1 \\ \Rightarrow \sum_{k=1}^n (k^3 - (k-1)^3) &= \sum_{k=1}^n (3k^2 - 3k + 1) \\ &\Rightarrow n^3 = 3 \sum_{k=1}^n k^2 - 3 \sum_{k=1}^n k + n \\ &\Rightarrow 3 \sum_{k=1}^n k^2 = n^3 + 3 \cdot \frac{n(n+1)}{2} - n \\ &\Rightarrow \sum_{k=1}^n k^2 = \frac{1}{3}(n^3 - n) + \frac{n(n+1)}{2} \\ &\Rightarrow \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}\end{aligned}$$

Q.10

Q.10 Derive the formula for $\sum_{k=1}^n k^3$.

Sol:

$$\begin{aligned}
k^4 - (k-1)^4 &= 4k^3 - 6k^2 + 4k - 1 \\
\Rightarrow \sum_{k=1}^n (k^4 - (k-1)^4) &= \sum_{k=1}^n (4k^3 - 6k^2 + 4k - 1) \\
&\Rightarrow n^4 = 4 \sum_{k=1}^n k^3 - 6 \sum_{k=1}^n k^2 + 4 \sum_{k=1}^n k - n \\
&\Rightarrow 4 \sum_{k=1}^n k^3 = n^4 + n(n+1)(2n+1) - 2n(n+1) + n \\
&\Rightarrow \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2
\end{aligned}$$

Q.11

Q.11 Show that a subset of a countable set is also countable.

PF:

Let B is countable, and $A \subseteq B$. Then there exists an injection $h : B \rightarrow \mathbb{N}$.

Then, let $i : A \hookrightarrow B$, i.e., $\forall a \in A, i(a) = a$.

Thus there exists an composed injection $h \circ i : A \rightarrow \mathbb{N}$, which means A is also countable.

Q.E.D..

Q.12

Q.12 Assume that $|S|$ denotes the cardinality of the set S . Show that if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$.

PF:

We have $|A| = |B|$ and $|B| = |C|$, then there exists bijections $f : A \rightarrow B$ and $g : B \rightarrow C$.

Thus for the composition $g \circ f : A \rightarrow C$:

Injectivity: If $(g \circ f)(a_1) = (g \circ f)(a_2)$, then by injectivity of g we have $f(a_1) = f(a_2)$, and by injectivity of f we get $a_1 = a_2$.

Surjectivity: For any $c \in C$, we have $b \in B$ s.t. $g(b) = c$ by the surjectivity of g , then we have $a \in A$ s.t. $f(a) = b$ for the surjectivity of f . Therefore $(g \circ f)(a) = c$.

Hence $g \circ f$ is a bijection $A \rightarrow C$, i.e., $|A| = |C|$.

Q.E.D..

Q.13

Q.13 If A is an uncountable set and B is a countable set, must $A - B$ be uncountable?

Sol: It **must be uncountable**.

Assuming that $A - B$ is countable. $A - B = A \cap \overline{B}$, then

$$A = (A \cap \overline{B}) \cup (A \cap B) = (A - B) \cup (A \cap B).$$

For $A \cap B \subseteq B$ and B is countable, $A \cap B$ is countable. And for the union of two countable sets is countable, A is also countable, which leads to a contradiction.

Therefore, $A - B$ is uncountable.

Q.14

Q.14 By the Schröder-Bernstein theorem, prove that $(0, 1)$ and $[0, 1]$ have the same cardinality.

Schröder-Bernstein: If there is an injection $A \rightarrow B$ and an injection $B \rightarrow A$, then there exists a bijection $A \leftrightarrow B$.

Injections:

- $i : (0, 1) \rightarrow [0, 1], i(x) = x$ (injection).

- $j : [0, 1] \rightarrow (0, 1)$, w.l.o.g., $j(x) = \frac{x+1}{3}$, for j is strictly increasing, it's obviously an injection.

By Schröder-Bernstein, there exists a bijection $(0, 1) \leftrightarrow [0, 1]$, therefore $|(0, 1)| = |[0, 1]|$.

Q.15

Q.15 If $f_1(x)$ and $f_2(x)$ are functions from the set of positive integers to the set of positive real numbers and $f_1(x)$ and $f_2(x)$ are both $\Theta(g(x))$, is $(f_1 - f_2)(x)$ also $\Theta(g(x))$? Either prove that it is or give a counter example.

Sol: **No**.

Counterexample: Let $g(x) = f_1(x) = f_2(x) = x$. Then obviously $f_1, f_2 \in \Theta(g)$ but $(f_1 - f_2)(x) \equiv 0$. Assuming that $0 \in \Theta(g)$, then there exist $c_1, c_2 > 0$ and N s.t. $c_1 \cdot g(x) \leq 0 \leq c_2 \cdot g(x)$ for all $x \geq N$, which is obviously impossible, which leads to a contradiction.

Thus $(f_1 - f_2) \notin \Theta(g)$.

Q.16

Q.16 Show that if $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_{n-1}$, and a_n are real numbers and $a_n \neq 0$, then $f(x)$ is $\Theta(x^n)$.

PF:

For $x \geq 1$, via triangle inequality: $|f(x)| \leq (|a_n| + \dots + |a_0|) x^n$. Thus let $c_2 = |a_n| + \dots + |a_0|$, we have $f(x) = O(x^n)$.

Plus, let $\alpha = |a_n|$ and $M = |a_{n-1}| + \dots + |a_0|$. For $x \geq 1$, $|a_{n-1} x^{n-1} + \dots + a_0| \leq M x^{n-1}$, thus, $|f(x)| \geq \alpha x^n - M x^{n-1} = x^{n-1}(\alpha x - M)$.

Let $N = \max\{1, \lceil 2M/\alpha \rceil\}$. For $x \geq N$, $\alpha x - M \geq \frac{\alpha}{2} x$, so $|f(x)| \geq (\alpha/2) x^n$. Then let $c_1 = \alpha/2$, we have $f(x) = \Omega(x^n)$.

Therefore, $f(x) = \Theta(x^n)$.

Q.E.D..

ExtraPF:

Via Limit Test:

We have $\lim_{x \rightarrow \infty} \frac{|f(x)|}{x^n} = |a_n| \in (0, \infty)$.

By the definition of limit, w.l.o.g., take $\varepsilon = |a_n|/2$, there exist $c_1, c_2 > 0$ and N such that $c_1 x^n \leq |f(x)| \leq c_2 x^n$ for all $x \geq N$. Hence $f(x) = \Theta(x^n)$.

Q.E.D..

Q.17

Q.17 Prove that $n \log n = \Theta(\log n!)$ for all positive integers n .

We have $n! \leq n^n$, then $\log n! \leq n \log n$.

Obviously we have $n! \geq (n/2)^{n/2}$, then $\log n! \geq \frac{n}{2} \log\left(\frac{n}{2}\right) = \frac{1}{2 \log 2} n \log n$.

Thus $\log n! = \Theta(n \log n)$.

Q.E.D..

Q.18

Q.18

- (a) Show that this algorithm determines the number of 1 bits in the bit string S :

Algorithm 1 bit count (S : bit string)

```
count := 0
while  $S \neq 0$  do
    count := count + 1
     $S := S \wedge (S - 1)$ 
end while
return count {count is the number of 1's in  $S$ }
```

Here $S - 1$ is the bit string obtained by changing the rightmost 1 bit of S to a 0 and all the 0 bits to the right of this to 1's. [Recall that $S \wedge (S - 1)$ is the bitwise *AND* of S and $S - 1$.]

- (b) How many bitwise *AND* operations are needed to find the number of 1 bits in a string S using the algorithm in part a)?

1. PF:

Loop invariant: After i -th iteration, count is i and the current S is the last S with its rightmost 1-bit removed.

Initialization: At the beginning, $i = 0$, and the count is 0, and it's trivially satisfied the loop invariant.

Maintenance: For $S - 1$, the rightmost 1-bit of S will be removed, and all the 0-bits at the rightside of the 1-bit will become 1-bits. Then for the bitwise *AND* operation, all the noticed bits will eventually become 0-bits. Which is to say, the rightmost 1-bit will be removed, which satisfy the loop invariant.

Termination: The loop will terminate when $S = 0$, i.e., there's no 1-bit in S , and the iterate times, which is the count, is the number of 1-bits.

Q.E.D..

2. Sol:

In each iteration, there will be a bitwise *AND* operation, and the rightmost 1-bit of S will be removed. Thus the number of bitwise *AND* operations equals to the number of **1-bits** in S .

Q.19

Q.19

- (1) Show that $(\sqrt{2})^{\log n} = O(\sqrt{n})$, where the base of the logarithm is 2.
- (2) Arrange the functions

$$n^n, (\log n)^2, n^{1.0001}, (1.0001)^n, 2\sqrt{\log_2 n}, n(\log n)^{1001}$$

in a list such that each function is big- O of the next function.

1. PF:

We have $a^{\log_b n} = n^{\log_b a}$, then $(\sqrt{2})^{\log_2 n} = n^{\log_2 \sqrt{2}} = n^{1/2} = \sqrt{n}$, thus, $(\sqrt{2})^{\log n} = \Theta(\sqrt{n})$, then $(\sqrt{2})^{\log n} = O(\sqrt{n})$ is trivially true.

2. Sol:

$$(\log n)^2 \prec 2\sqrt{\log_2 n} \prec n(\log n)^{1001} \prec n^{1.0001} \prec (1.0001)^n \prec n^n$$

Q.20

Q.20 Give an example of two increasing functions $f(n)$ and $g(n)$ from the set of positive integers to the set of positive integers such that neither $f(n)$ is $O(g(n))$ nor $g(n)$ is $O(f(n))$.

Sol:

Let $a_k = 2^{2^k}$ and blocks $B_k = [a_k, a_{k+1}) \cap \mathbb{N}$.

Let:

$$f(n) = \begin{cases} n^2, & n \in B_k, k \text{ is even} \\ a_k n, & n \in B_k, k \text{ is odd} \end{cases} \quad g(n) = \begin{cases} a_k n, & n \in B_k, k \text{ is even} \\ n^2, & n \in B_k, k \text{ is odd} \end{cases}$$

Inside any block both n^2 and $a_k n$ increase in n .

At boundaries, $f(a_{k+1} - 1) < f(a_{k+1}) = a_{k+1}^2$ and $g(a_{k+1} - 1) < g(a_{k+1}) = a_{k+1}^2$, then for $a_{k+1} \gg a_k$, f, g are both increasing.

On even blocks, let $n = a_{k+1} - 1$, $f(n)/g(n) = n/a_k \geq (a_{k+1} - 1)/a_k \rightarrow \infty$, thus $f \neq O(g)$.

On odd blocks, let $n = a_{k+1} - 1$, $g(n)/f(n) = n/a_k \geq (a_{k+1} - 1)/a_k \rightarrow \infty$, thus $g \neq O(f)$.

Which satisfy the statement.

Q.21

Q.21 Aliens from another world come to the Earth and tell us that the 3SAT problem is *solvable* in $O(n^8)$ time. Which of the following statements follow as a consequence?

- A. All NP-Complete problems are solvable in polynomial time.
- B. All NP-Complete problems are solvable in $O(n^8)$ time.
- C. All problems in NP, even those that are not NP-Complete, are solvable in polynomial time.
- D. No NP-Complete problem can be solved *faster* than in $O(n^8)$ in the worst case.
- E. $P = NP$.

Sol:

- A. **True.** For all NP-Complete problem X , there is a poly-time reduction $X \leq_p 3SAT$. Compose it with the $O(n^8)$ 3SAT solver to get a polynomial-time algorithm for X .
- B. **False.** The reduction may map size n to $m = \text{poly}(n)$. Then the runtime becomes $O(m^8) = O(n^{8k})$ for some $k \geq 1$, not necessarily $O(n^8)$.
- C. **True.** If an NP-Complete problem (like 3SAT) is solvable in polynomial time, then $P = NP$, thus all problems in NP are solvable in polynomial time.
- D. **False.** An $O(n^8)$ algorithm for 3SAT doesn't give the worst-case lower bound for other NP-Complete problems, then it might be faster.
- E. **True.** 3SAT is NP-Complete and has a polynomial-time algorithm, we then have $P = NP$.

Q.22

Q.22 Compare the following pairs of functions in terms of order of growth. In each of the following, determine if $f(n) = O(g(n))$, $f(n) = \Omega(g(n))$, $f(n) = \Theta(g(n))$. There is **no need** to explain your answers.

$f(n)$	$g(n)$	
(1) $(\log_2 n)^a$	n^b	here $a, b > 0$
(2) $2^{n \log_2 n}$	$10n!$	
(3) $\sqrt{n}$	$(\log_2 n)^5$	
(4) $\frac{n^2}{\log_2 n}$	$(n \log_2 n)^4$	
(5) $\log_2 n$	$\log_2(66n)$	
(6) $1000(\log_2 n)^{0.9999}$	$(\log_2 n)^{1.001}$	
(7) n^2	$n(\log_2 n)^{15}$	

Sol:

1. $f(n) = O(g(n))$.
2. $f(n) = \Omega(g(n))$.
3. $f(n) = \Omega(g(n))$.
4. $f(n) = O(g(n))$.
5. $f(n) = \Theta(g(n))$.
6. $f(n) = O(g(n))$.
7. $f(n) = \Omega(g(n))$.